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Language variation

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In this chapter, we continue to discuss the language architecture and, more specifically, parameters throughout the history of the generative enterprise. This time around, we will take a chronological approach and describe how the concept of a parameter has changed over the years and what the current idea of a parameter is. We will keep in mind how these ideas influenced the research questions of L2 acquisition studies performed since the 80s and what we can expect for the future. We will constantly keep an eye on how this type of theoretical knowledge can improve classroom teaching efficiency. But first, we will look at a few examples of language differences.

2.1 How do languages differ?

Before we go into a discussion of how differences among languages have been captured, analyzed, and conceptualized, it is important to get a good idea of how it is that languages vary. The most obvious difference, of course, is in their lexicons: different languages have different words for the same entity. Take for example the English word *table*, the Bulgarian word *masa*, the Russian word *stol*, and the Vietnamese word *bảng*, all standing for a

piece of furniture with a flat top and one or more legs, providing a level surface on which objects may be placed. All the words in a second language have to be learned, usually painstakingly and one by one, but there are shortcuts and regularities to that process as well, as we shall see in Chapter 9 on acquisition of the lexicon. Languages also differ in what sounds they employ and how they combine them, the so-called phonotactic constraints. For example, a sequence of the sounds /t/+// is unpronounceable in English at the beginning of a word, but perfectly fine in Bulgarian, as in the word *tləst* ‘fat.’

Abstracting away from the lexicon and the phonetics/phonology, however, syntactic variation is not random, and that is clearly brought home by extensive work on linguistic typology, stemming from Greenberg’s (1963) pioneering work on language universals. Describing the possible order possibilities between the main constituents of the sentence: S(subject)–O(object)–V(verb), for example, linguists have established that not all of the six possible orders appear with the same frequency among languages of the world. If you want to discover such regularities on your own, please refer to the *World Atlas of Language Structures*.¹ (However, be forewarned that this is an extremely addictive website.) There you can find a map of the world with the possible word orders marked clearly. Out of 1377 languages described, 565 utilize SOV (41%), 488—SVO (35.4%), and 95—VSO, while 189 languages display no dominant order.

What these numbers suggest, first of all, is that the order of the major sentence constituents cannot be described in absolute universal terms. Rather, we are dealing with universal statistical tendencies: SOV and SVO are the most frequent word orders, etc. Furthermore, and much more interesting from an acquisition point of view, there are certain other regularities that go together with the dominant word order. But before we look at these regularities, we need to brush up on the notions of “head” and “complement” of a phrase. The head is the most important word in the phrase, from a structural as well as meaning perspective. The head determines the category of the whole phrase. The complement represents additional information that completes the meaning. The noun (underlined) is the head of the noun phrase (e.g., *the leg of the table*); the verb is the head of the verb phrase (*see the leg of the table*), etc. Languages that place heads either at the end or at the beginning of the phrase are called *harmonic*. For example, if a language has O before V, it is likely that the language will also

¹ WALS, <http://wals.info/cf>. Haspelmath et al. (2005).

have postpositions (*the market to*); if a language has V then O, it is highly likely that it will have prepositions (*to the market*).² In fact, 94.5% of languages surveyed in the WALS are harmonic with respect to the order of V, O, and adpositions (Baker 2008). Other word order regularities that go together with the order of V and O are given below (Biberauer 2008):

- (1) If V–O, then Auxiliary–V, Prepositions, Noun–Adjective, Noun–Genitive, Noun–Relative Clause.
If O–V, then V–Auxiliary, Postpositions, Adjective–Noun, Genitive–Noun, Relative Clause–Noun.

Now let us take a concrete example from Japanese and English, from Baker (2003). Notice that linguistic examples are given on three lines, the first line providing an approximate transliteration of the Japanese sounds, but already divided into grammatical and lexical morphemes. The second line gives English equivalents of these morphemes, called *glosses*, retaining their order in Japanese. The third line gives an idiomatic translation of the sentence meaning in English. For our linguistic analysis purposes now, pay attention to the second line first.

- (2) Taro is thinking that Hiro showed pictures of himself to Hanako.
- (3) Taroo-ga [Hiro-ga Hanako-ni zibun-no syasin-o miseta]
Taro Hiro Hanako-to self-of picture showed
to omotte iru.
that thinking is
‘Taro is thinking that Hiro showed pictures of himself to Hanako.’

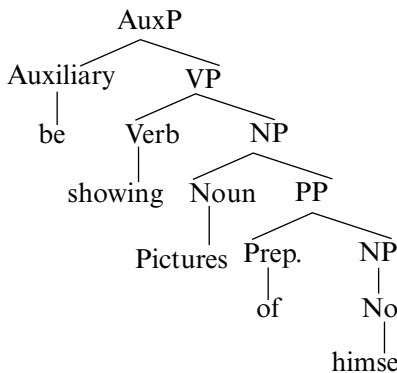
At first glance, it may seem that the two sentences in (2) and (3) are very different. In fact, if you read out the second line of (3), it sounds like gibberish. However, if you look closely, you can see that the typological generalizations given in (1) above are exemplified clearly in these two examples. English is V–O while Japanese is O–V. The auxiliary *is* precedes the verb *thinking* in English, while the auxiliary *iru* ‘is’ follows the verb *omotte* ‘thinking’ in Japanese. The English preposition goes in front of the noun phrase (NP) to make a prepositional phrase (PP) *of himself*, while in Japanese the postposition *-no* ‘of’ follows the NP (*himself of*).

² Prepositions and postpositions have a common label: they are called “adpositions.”

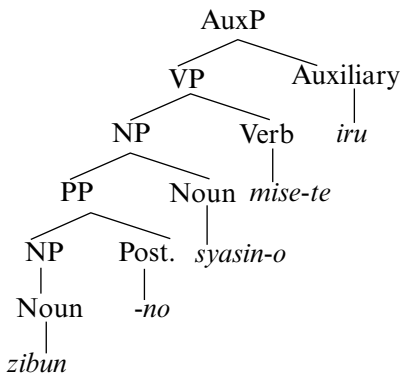
Before going forward, find two other differences between English and Japanese that can be attributed to their word order difference. Hint: there is no relative clause in this example, but there is a subordinate clause that functions as an object.

More generally speaking, English seems to put new words (the heads) in the beginning of already constructed phrases (acting as complements), whereas Japanese seems to add new words at the end of phrases. If we have a look at Baker's simplified phrase structure trees of parts of these two sentences, we will see why. Note how the two phrase structures look like mirror images of each other. Secondly, if you compare line for line, you will see the opposite order of the constituents: Verb and Object NP, Preposition/Postposition and NP, etc. One way to express this generalization is to say that English phrases are predominantly *head-initial* (the head comes before the complement), while Japanese phrases are predominantly *head-final*. This is known as the Head Directionality Parameter (Greenberg 1963, Stowell 1981, Dryer 1992).

(4) English phrases (partial tree)



Japanese phrases (partial tree)



Next, to illustrate another level of language variation, we shall take a second paradigmatic distinction from Baker (2003), based on his work on the Mohawk language. Mohawk is a polysynthetic language, which means that “its sentences seem simple and fluid, but word structures are complex and rigidly structured” (Mithun and Chafe 1979, Baker 1996). Let us compare the two Mohawk sentences below with their English translations, the third line in the examples (5) and (6).

- (5) Rukwe' wa-sh-ako-hsir-u ne owira'a
man past-he-her-blanket-gave the baby
'The man gave the baby a blanket.'
- (6) Rukwe' wa-h-uwa-hsir-u ne owira'a
man past-him-she-blanket-gave the baby
'The baby gave the man a blanket.'

Let us start from the translations this time. Clearly, the meanings of the two sentences are exactly the opposite in terms of who is performing the action of giving and who is the recipient. Next, focus on the second line of the examples and notice that the word *rukwe'* 'man' is before the verb in both examples, but in one sentence he is the doer of the action while in the other he is the recipient. This fact illustrates the free word order in Mohawk, and it is very different from English, isn't it? In English, *John kissed Mary* and *Mary kissed John* do not denote the same event. Unlike English, the words in a Mohawk sentence can be put in any order without affecting the basic meaning. Now, look at the rich information encoded on the verb, in the form of tense and agreement morphemes. In (5), there is a morpheme showing that the event is past, then another morpheme pointing to the subject (3rd person, singular, masculine), and another indicating that the direct object is 3rd person, singular, feminine. Right before the verb, you see that the whole object *hsir* 'blanket' is "incorporated" into the verb. If you now compare the second and the third morphemes on the verb in (6), they indicate that this time the recipient is masculine while the giver is feminine, hence the different meaning. Mohawk speakers can also drop the nouns in these sentences altogether, if they are already mentioned in the discourse or known to the interlocutors. The event of giving in both languages has three participants: the giver, the recipient, and the object that changes hands, all three have to be obligatorily expressed. Baker proposes that the parametric difference between English and Mohawk can be formulated like this:

- (7) Mohawk: Express every major participant in an event inside the verb that names the event.

English: Don't express the participants in an event inside the verb that names the event.

Apart from this major difference, visible in every sentence, note that there are similarities between the two languages, too. The obligatory nature of the three event participants is one similarity; the constraint that only the direct object, but no other participant, can be incorporated into the verb is another similarity. Although English doesn't have noun incorporation, it has verbal compounds such as *gift-giving*, while *baby-giving* (even if attested) can only have the meaning that the baby is the object that changes hands, but cannot be the recipient or the giver.

Teaching relevance

These are just two examples of how human languages differ parametrically, or in a predetermined way. We are interested in differences between languages, because they are what second language learners have to acquire. Furthermore, we would like to be able to describe those differences in a systematic way that makes acquisition easy. For example, if an English native speaker is acquiring Japanese, she does not need to acquire all the typological generalizations one by one. Having acquired one of them, for example O precedes V, she will be able to hypothesize that Japanese is a head-final language and has postpositions, etc. If an English native speaker is acquiring Mohawk, she will expect that in every sentence, the event participants have to be marked on the verb. Such generalizations help language acquisition immensely.³ There is empirical support for such generalizations in the work of Megan Smith and Bill VanPatten (2014).⁴

2.2 Principles and Parameters in history

In this section, we will take a chronological survey of the concept of parameters. In doing so, we are aiming to elucidate how the concept of parameter developed, and why it matters for SLA. During the 1960s and

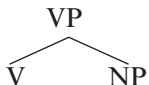
³ This generalization was tested at the outset of GenSLA research in the 1980s, in the seminal work of Suzanne Flynn (Flynn 1987a,b, Flynn and Espinal 1985).

⁴ The researchers tested native English speakers, who had never studied Japanese before. At first, participants were exposed to some head-final Japanese phrases, and then tested them on congruous phrases that they had not seen in the training part of the study. After minimal exposure, the participants demonstrated awareness of the head-final nature of Japanese, as measured by longer reading times on ungrammatical sentences as compared to the grammatical sentences.

70s, the attention of generative linguists focused on the basic regularities of language structure and on how to capture the infinite surface realizations of the underlying finite structure. Attention was paid to fine-grained descriptions of individual languages in rewrite rules⁵ (as well as transformations), as exemplified in (8), where the order of the constituents is important:

(8) $NP \rightarrow \text{Det } N$ (as in *the dog*); $VP \rightarrow V \text{ } NP$ (as in *eat a melon*), etc.

These rewrite rules are to some extent notational variants of tree structures, where the arrow stands for the binary branching. They both symbolize a VP that immediately contains a V and an NP. The order of V and NP captures the fact that the verb precedes its object (as in English).

(9) $VP \rightarrow VNP$ = 

Note the similarity with the English tree and the difference with the Japanese tree above in (4). To the extent that the grammar of individual languages was fully accounted for by such rules, it was said that the grammar was *descriptively adequate*.

Over time, descriptive rules proliferated, which was in conflict with another stated goal of the generative framework, that of *explanatory adequacy*. The latter attempted to elucidate how a child was able to acquire language, in such a short time and relatively error-free, on the basis of finite (and maybe limited) comprehensible input. The abundance of descriptive rules describing language specific constructions was difficult to square with the easy, effortless way children learn language. In short, the system of rules became too rich to explain language acquisition.

This inherent conflict between descriptive and explanatory adequacy brought about the Principles and Parameters approach (P&P) in the beginning of the 80s. Explanatory adequacy came to be of higher priority and more highly valued than descriptive adequacy.

⁵ Rewrite rules are called that because what is on the left of the arrow gets rewritten on the right-hand side of the arrow.

Since solving the clash between descriptive and explanatory adequacy (satisfying both) was a major goal of the Principles and Parameters approach, linguists developed an elaborate but elegant technical apparatus to this end. The points of invariance and points of variation were unified by proposing parameterized (flexible) principles. The transition from rules to principles was complete.

From the perspective of SLA, there is nothing wrong with specific language rules, and that is how language instruction has traditionally proceeded. However, explanatorily adequate parameters offer a shortcut of added efficiency to the instruction and acquisition process. Furthermore, this emphasis on explanatory adequacy revealed linguistic properties that were part of the knowledge of language, but were not within educators' purview until that time. The language gates had opened.

We shall exemplify a parameterized principle with Subjacency, one of the earliest proposals that was also taken up and studied extensively by second language scholars. Chomsky (1973) proposed that the leftward movement of a *wh*-constituent (such as *what*, *where*, or *which student*), starting from its underlying position close to its related verb and moving to the initial position of the sentence, is constrained by the following metric. Such a constituent could move only in small steps that do not cross over more than one "bounding node." Bounding nodes in the initial proposal were the categories S (standing for sentence) and NP. Let us look at a declarative sentence first, such as the one in (10), where the object *the present* and the location *under the bed* are close to the verb *put*. Sentence (11) questions the location; the question word *where* has moved to the top of the embedded clause. Sentence (11a) can be analyzed as a result of the two steps illustrated in (11b) and (11c). First the location phrase is substituted with the question word *where*, and then it moves to the beginning of the embedded clause leaving a trace, or a copy of itself, marked with angled brackets. The lower copy is silent, but it is there. In this jump, the *wh*-phrase moves over one bounding node, the S, which makes the jump a legitimate one.⁶

(10) Bill put the present under the bed.

⁶ The NP *the present* was considered to be a sister node to the location phrase *where* on the tree, so it is not counted as being "jumped over."

what from stopping over in that same position on its way to the top of the sentence. If *what* could stop over in the position of *where*, it would, but it cannot since that position is already taken (the node is filled). Remember that this position is considered the escape hatch from the island. The sentence in (12) is unacceptable because the question word *what* is forced to make an illegitimately long jump to the top of the main clause; the sentence in (13) is acceptable because *what* is allowed to stop over half-way, leave a copy of itself, and continue on its way to the top of the sentence, thereby making two smaller jumps. Note that now we have two lower copies of *what*, neither of which are pronounced. But if the top copy of *what* in the escape hatch did not exist, we would not be able to explain the difference in acceptability between (12a) and (13a).

Do an informal linguistic experiment by writing out sentences (10), (11), (12), and (13) on a piece of paper, and poll three (or more) of your friends about the acceptability of the sentences. To start with, ask whether they are acceptable or unacceptable. Then offer your “experimental participants” a scale of acceptability, say 1 to 5 or 1 to 7, where 1 is unacceptable and 7 is perfect. Long sentences with *wh*-movement like these ones are never judged perfectly acceptable, because they are difficult to process. That is why I predict that example (10) will obtain higher ratings. See whether you will be able to confirm the lower acceptability of (12) in comparison with (11) and (13).

The reader has to appreciate the breathtaking elegance and attractiveness of this analysis. This was precisely the type of data that the generative program was called upon to reveal and explain. It uncovered linguistic knowledge that speakers did not even suspect that they had. This data set and others like it embodied the inherent complexity and surface unpredictability of the linguistic input children were exposed to. Very often, analogy between possible linguistic changes in sentences is misleading (think of the paradigm in (11), (12), (13) above, among many other examples). At the same time, such data sets become part of children’s grammatical knowledge easily. The argument from the Poverty of the Stimulus goes like this:

1. English native speakers generally agree that (11) and (13) are acceptable sentences, so the rules of generating these sentences are part of their grammar.

2. However, nobody teaches English children that sentences such as (12) are unacceptable. Nobody *can* teach English children that, because it is not part of speakers' explicit language knowledge.
3. Since children know the difference between (11) and (13) on the one hand, and (12) on the other, they derive this knowledge from the innate Language Acquisition Device, or Universal Grammar.

Let me unpack step 3 of the argument a little. It is conceivable that children hear sentences such as (11) and (13), but not (12), since the latter is ungrammatical and no one around them will pronounce it. Could it be that children capitalize on never having heard a sentence like (12) and generalize from the lack of evidence ("I have never heard this sentence") to evidence of lack ("this sentence is unacceptable in my language")? But there are many sentences that children have never heard, yet they produce them. The generalization going from lack of evidence to evidence of lack is fallacious. It must be the case, then, that children know more than they are exposed to. They know that some structures are illicit in their language. The explanation of this fact must lie in Universal Grammar, which has supplied children with this linguistic information.

The history of describing language variation continues. Subjacency was treated as a principle of UG, in the sense that no language allowed movement of a *wh*-phrase to cross over more than one bounding node. However, uncovering language variation, Rizzi (1982) proposed that Italian has different bounding nodes than English. In the clause structure, the S node is dominated by S-prime (S', or what we currently know as complementizer phrase, or CP). Rizzi proposed that not S but S' was a bounding node in Italian. This subtle difference allowed sentences of the type of (12) to be acceptable in Italian. It was also discovered that Russian was even more restrictive than English and Italian (Freidin and Quicoli 1989), and so S and S' were proposed to be bounding nodes. Languages that did not move their *wh*-phrases to the beginning of the clause, such as Mandarin Chinese, were considered not to exhibit Subjacency at all. Thus phrase structure and movement were both subject to the inviolable principle, Subjacency, but the precise manner in which this principle manifested itself in different languages was flexible, and constrained by parametric variation.

Not all principles were flexible and parameterized. For example, the Extended Projection Principle, which stipulated that every clause should have a subject, and the Theta Criterion, which postulated that each thematic

role of a verb should be assigned to one and only one argument, were inviolable and universal. Subjacency, too, was initially thought to be an invariant principle (cf. Chomsky, 1973), but Rizzi's (1982) work revealed this assumption to be incorrect. Thus the Principles and Parameters framework was very much a research program, where accepted analyses were likely to change a few years later, with the discovery of exciting new data from fresh languages. As in normal scientific practice, this was considered a strength, not a weakness of the framework.

A second, and very important characteristic of the P&P style parameters was the notion of parameter clusters. The gist of this idea was Chomsky's famous metaphor, attributed to James Higginbotham, of parameters as light switches with two values: On and Off. Placing the switch in one position leads to a cascade-like effect with various other constructions becoming available in the grammar of a speaker. This was known as the *clustering effect*. The notion was conceptually attractive because the constructions in the cluster were superficially unrelated. If it was possible to show that they all depended on an underlying value of a single parameter, this would constitute a cogent argument for the existence of UG indeed. Based on historical evidence, Lightfoot (1993) proposed the condition that the parametric *trigger*, the cue for the setting of the whole parameter, had to be salient and easy to acquire, such as a piece of inflectional morphology.

We will exemplify this idea of a parameter cluster with the Null Subject Parameter already introduced in Chapter 1. It is a classic parameter, which held a lot of promise at the time. In languages such as Spanish and Italian, it is possible not to pronounce the subject of a clause when the information of who the doer of the action is comes from the agreement morphology on the verb, as shown in (14a). This is not so in English, French, and German, see the English translations of the examples below to compare.

- (14) a. *Habl-a español* (Spanish)
 speak-3SG Spanish
 'She/he speaks Spanish.'
- b. *Piov-e* (Italian)
 rain-3SG
 'It is raining.'
- c. *Verrà Gianni*
 will-come-3SG Gianni
 'Gianni will come. / *Will come Gianni.'

- d. Chi cred-i che <chi> verrà?
 who believe-2SG that <who> will come
 ‘Who do you think that will come?’ (ungrammatical in English)

The proposed cluster of constructions included null expletives (dummy subjects) as in (14b), postverbal subjects as in (14c), a sequence of the complementizer *that* and a trace of the moved subject, known as the “*that*-trace effect” as in (14d), and rich subject–verb agreement (Rizzi 1982). The salient cue that is observable from the linguistic input is the unrealized pronominal subject. The constructions in the cluster appear to be superficially unrelated and of varying difficulty. However, all of them hinge on the availability of a null pronominal subject, let’s call it *pro* (pronounced “small *pro*” or “little *pro*”). Recall the universal principle that every sentence must have a subject. This requirement is still satisfied in the sentences in (14), but by *pro*. This null subject appeared in “weather sentences” (14b) as the equivalent of overt *it* in English (*It rains.*) *Pro* allows the notional subject *Gianni* in (14c) to remain lower in the structure, creating this postverbal subject effect. Finally *pro* also helps the *that*-trace violation to be avoided, as in (14d), by allowing *wh*-subjects to be extracted from a lower position, while *pro* itself once again satisfies the subject requirement. The “rich” subject–verb agreement in Spanish and Italian supplies *pro* with the person and number features it intrinsically lacks.

The details of some of these analyses have changed over time; however, the basic point still remains: the constructions exemplified in (14) cluster in null-subject languages, while their exact opposites are acceptable in non-null-subject languages. The parameter’s two values hinge on the existence or unavailability of *pro*: it is the basis of the underlying analyses of the whole cluster. The rich deductive structure of this parameter was the poster child in the quest for explanatory adequacy. Null pronominal subjects are relatively easy to notice in the primary linguistic data; and although *that*-trace constructions are rare and not so salient, they would come into the grammar for free (when all the other necessary linguistic properties are learned).

A significantly bigger challenge is acquiring the fact that *that*-trace sentences are NOT part of the grammar of non-null-subject languages such as English. The child is not going to hear the unacceptable sentence in (15) and others like it. However, the easy analogy between sentences such as (16) and (17) below suggests that (15) should be fine. No parent teaches children that extraction of a subject after an overt complementizer *that* is unacceptable,

because this knowledge is not part of the explicit language knowledge speakers of English possess.

(15) *Who do you think that will come?

(16) Who do you think will come?

(17) I know that he will come.

Knowledge of the unavailability of (15) is another case of the Poverty of the Stimulus. In this instance, knowledge will be supplied by the acquisition that *pro* is not part of the English grammar. Noticing overt pronominal subjects and especially dummy subjects *it* and *there* would lead the child to acquire the whole cluster. Keeping all of this in mind, one could predict that a child that had already acquired the Null Subject Parameter would gain knowledge of the whole cluster.

As Baker (2008: 352) noted, history has not been kind to the Null Subject Parameter as originally stated by Rizzi (1982). The parameter leads to various expectations: that null-subject languages will be morphologically rich, that morphologically poor languages will not be null-subject languages, that the properties exemplified in (14) will always cluster together, and so on. It was soon discovered that these predictions were not all borne out. For example, Icelandic and Russian are rich in verbal agreement morphology, but are not null-subject languages. One attempted remedy was to argue that the setting of some other parameter interfered with the Null Subject Parameter. Icelandic was a case in point, where the setting of the verb-second parameter was proposed to mask the effects of null subjects, but that explanation cannot be extended to Russian. Another research direction was to quantify exactly how rich “rich agreement morphology” has to be in order to license null subjects. Both attempts were ultimately unsuccessful. However, both were nice tries, and scientists can only be faulted for not trying to explain existing states of knowledge, not for their explanations eventually failing.

More recently, Snyder (2001) has proposed another parametric cluster that has better withstood the test of time and extended scrutiny. The Compounding Parameter unifies (at least) the following constructions in a parametric cluster: productive noun–noun compounds as in (18), the verb–particle construction (19), the double object construction (20), resultative predicates (21), and *make*-causatives (22). The property underlying all these constructions has to do with the availability of complex predicate constructions, commonly involving productive root-plus-root compounding:

- (18) *spider box* (meaning: a box where spiders are kept, a box with a spider drawn on its lid, etc.), *soda lady* (meaning: a lady who sips soda, a lady who sells soda, a lady who is going to a soda convention, etc.)
- (19) Mary ate the sandwich up.
- (20) Mary gave John a rose garden.
- (21) John wiped the table clean.
- (22) Mary made John leave.

From its very conception, this parameter unified child language acquisition data and typological observations. Snyder observed a one-way implication in the data from a sizable number of languages: If a language permits the verb-particle construction, then it also allows free creation of novel compounds like *spider box* or *soda lady*. The implication is unidirectional, however: There do exist languages that allow this type of compounding, yet lack the verb-particle construction. Snyder therefore proposed that the grammatical pre-requisite for the English type of compounding (i.e., the positive setting of the compounding parameter) is one of several prerequisites for the verb-particle construction. A clear acquisition prediction can be made: Any given child acquiring English will either acquire compounding first, or acquire compounding and the verb-particle construction at the same time. In no case will a child acquire the verb-particle construction significantly earlier than compounding. This prediction received strong support from the longitudinal production data of twelve children (Snyder 1995, 2001).

2.3 The Minimalist Program

As illustrated in the previous section with some representative examples, a number of prominent parameters from the 1970s and the 80s met with the harsh reality of additional linguistic data. This situation led some researchers to reject the usefulness of the parametric endeavor as a whole and to question whether variation is really constrained (see, e.g., Newmeyer 2005). In this section, we will examine how the concept of parameters has changed, partly under the pressure of new data, partly because of a new phase in the Principles and Parameters enterprise, namely, the Minimalist Program.

The Minimalist Program, and especially its conceptualization after 2000, radically changed the approach to descriptive and explanatory adequacy, aiming to go “beyond explanatory adequacy,” which is the title of a programmatic paper by Chomsky (2004). The starting point of the Minimalist model is that the language faculty has to do its job in the most efficient manner possible, given the language architecture we discussed in Chapter 1. This approach in effect postulates that we have to start with the null hypothesis that language is “perfect” or “optimally designed,” at least until we get incontrovertible evidence to the contrary. Chomsky’s rationale was that research within the Principles and Parameters program had explained enough about the language faculty by the early 1990s, so that a new programmatic goal was possible that would drive scientific inquiry forward and answer novel research questions, such as, how did language evolve.

Even within the P&P analyses, economy of derivation, or the need for derivations to be maximally simple, had been used as a metric to evaluate them. The new Minimalist goal goes further: the number of purely language internal entities the theory postulates should be minimal. The simplified architecture comprises just the Lexicon, the Computational System (also known as the narrow syntax) and two interface components, PF and LF (see Chapter 1). The linguistic information flows from the narrow syntax to the two interfaces. Chomsky considers the operations at the interfaces to be of “virtual conceptual necessity,” because there is no getting away from the fact that language is a linearized sequence of symbols (words making up sentences and discourse) coupled with their interpretation. Thus PF and LF are conceptually necessary because they interface with the Articulatory-Perceptual system (the sound or signs) and with the Conceptual-Intentional system (the meaning).

Now, the requirement that language is as perfect as possible leads to the (strong) assumption that the computational system is the same for all languages, that is, it is universal. For example, it cannot be the case that the recursive operation Merge,⁸ the principal structure-building operation of the syntax, takes two syntactic objects in Language A and three syntactic objects in Language B. Merge always involves two objects (or nodes). Hence, parametric differences can no longer be captured by postulating

⁸ “Recursive” means that the operation happens again and again, as many times as needed.

that the syntax of two languages differs. Syntactic operations are universal. Note that this leaves us with only three places of possible variation: the lexicon and the two interfaces. We shall look at these possibilities in turn. Linguists' efforts are currently concentrated on reconceptualizing, or reformulating, the "old" parameters in the new Minimalist spirit, using the simplified language architecture.

The proposal that all parameters are in the lexicon because they are associated with the grammatical features of functional morphology is not new: it was made by Borer (1984) and adopted by Chomsky later on. This idea has been labeled the Borer–Chomsky Conjecture (Baker 2008: 253). It is currently a core assumption in Minimalist theorizing and one of the most influential proposals in the ongoing debate on where language variation is located. All parameters of variation are attributable to differences in the features of particular lexical items, the heads of functional categories, which reside in the lexicon. It is not difficult to see the conceptual appeal of this proposal. If lexical items constitute the sole locus of parametric variation, then learning the vocabulary of a first or second language constitutes learning the parametric profile of that language. In the next section, we will encounter the Bottleneck Hypothesis, a concrete proposal to this effect for L2 acquisition. We will also elucidate what we mean when we say "functional features."

Reconsideration of all the parameters proposed during the 30 years of generative theory development is far from complete, and some parameters are better candidates for reformulation than others. For example, the Null Subject Parameter, in effect, specifies two lexical requirements: the subject–verb agreement morphology in a null-subject language has to be sufficiently rich, and a particular type of pronoun, null little *pro*, must exist in the lexicon. The requirements of the compounding parameter can also be formulated as properties of lexical heads. Let me point out just one advantage of this idea. If variation is relegated to the lexicon, then variety is restricted to easily observable properties of utterances, hence easily learned.

We will next look at proposals that language variation can be captured at the PF and LF interfaces. In order to understand how this might work, we need to look more closely at features and where in the language architecture they are manifested.⁹ Features are properties of morphemes and words.

⁹ The following discussion is based on Gallego (2011).

They can be of three types: *phonological features* such as [$\pm$ sonorant], [$\pm$ voice], building up the sound side of the linguistic signal; *semantic features* such as [$\pm$ specific], [$\pm$ definite], [$\pm$ countable] for nouns and nominal phrases, [$\pm$ stative] [$\pm$ telic] for verbs and verbal phrases, etc.; and *formal morphosyntactic features* such as [Case: {Nom, Acc, Gen...}], [Person: {1, 2, 3}], [Number: {Sg, Pl}].¹⁰

But which features are involved in variation? An assumption shared by Chomsky and Jackendoff, for example, is that semantic features are not a source of variation, because all languages have to be able to express the same meanings. Formal features are also present in all languages, but variation is restricted to their actual manifestation. Hence, variation is down to the phonological features of the grammatical morphology, or to make this more precise, to the morpho phonological *expressions* of the syntactic and semantic features. Let me just reiterate this because it is very important: features and their expressions are different things. According to one approach, Halle and Marantz' (1993) Distributed Morphology, the latter are added to the linguistic message after the narrow syntax has produced a detailed phrase structure tree with all formal syntactic features on it evaluated, checked, and some of them deleted (more on this mechanism later on in the book) and the message has bifurcated into PF and LF interfaces.

One suggestion for capturing variation at PF goes as follows. At the end of the syntactic computation, the product is a syntactic (phrase structure) tree. Let us imagine that one item in this tree has moved from one position to another. We already described such movement when talking about Subjacency. The moved item is a copy of the initial item. In effect, after movement, we have two identical copies in two positions of the tree. One way this movement can be captured in the lexicon is to postulate that the attracting head has a feature that initiates the movement. An alternative way could be to think in terms of PF-features that determine which copy of a chain is ultimately pronounced. Some languages would choose to pronounce the top copy while others would choose to pronounce the bottom copy (Bobaljik 2002). In still other cases, which copy is pronounced may depend on the phonological makeup of the surrounding words (Bošković 2001). That would constitute variation at the PF interface. I am repeating

¹⁰ See the following website and Adger and Svenonius (2011) for more advanced discussion: <http://www.features.surrey.ac.uk/index.html>.

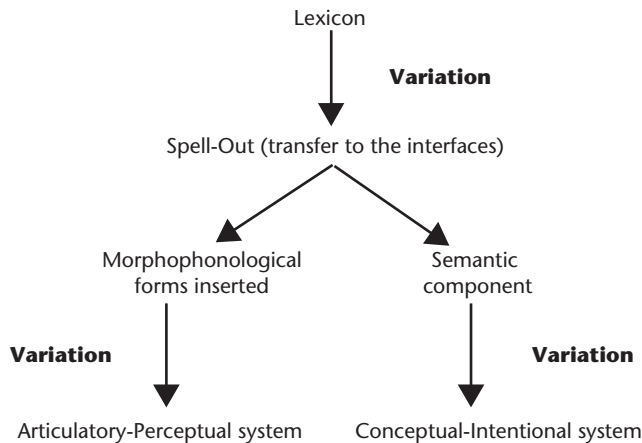


Figure 2.1 Minimalist Language Architecture

here the language architecture from Chapter 1, so that the reader can visualize our discussion better (Figure 2.1).

Other researchers have proposed that variation is possible at the LF interface, as well. For example, Ramchand and Svenonius (2008) argue for the possibility that the conceptual systems are universal, but that the mapping between syntax and the C-I systems is non-trivial. More specifically, they propose that certain components of meaning, such as Tense and Aspect, are universally required to be represented in the syntax, while certain other meanings, such as definiteness and specificity, may be supplied by the context. The former constitute the backbone of the sentence. The functional category Aspect is necessary to link the verb with its arguments (e.g., *eat* and *a sandwich*) and give the shape of the event encoded in the verb phrase (ongoing, complete, etc.). The functional category Tense makes sure that the event (e.g., *eat a sandwich*) is linked to a time of utterance (*Mary ate a sandwich* = past, *Mary will eat a sandwich* = future). The entire sequence of functional categories must always be present (at the syntax–semantics interface) to ensure that the C-I system contains the basic skeleton of the sentence meaning.

On the other hand, definiteness is a meaning that is expressed by a dedicated morpheme in some languages (English *the*, *a*) but is not overtly expressed in some other languages (Mandarin Chinese, Russian, Hindi). How do speakers of these languages understand that a certain noun phrase is to be interpreted as definite? That meaning can be supplied by the

context, or by other lexical means such as demonstratives. Other meanings that languages sometimes leave for the context to fill in are specificity, evidentiality,¹¹ and number. The interface, then, maps the information presented to it by the syntax and the information coming from the context. The actual variation is again captured in lexical items, but is also restricted by the requirements of the conceptual system.

Since different languages may employ different lexical items with different features to encode the same meaning, a crucial trait of this view of the grammar is underspecification, both in lexical and morpholexical (functional) morphemes. To take an example from Ramchand and Svenonius (2008), compared to the explicit dual marking in Northern Sámi personal pronouns *mii* ‘I and others’ and *moai* ‘I and one other’, English *we* is underspecified and can be disambiguated (but only if necessary) by adding lexical items (*I and others* versus *I and one other person*). Most of the time the context provides the missing information.

Finally, we will look at another proposal for the locus of variation, which is compatible with the conjecture that all variation is in the lexicon. The gist of this idea was proposed by Giorgi and Pianesi (1996), and it is currently assumed by many researchers. This type of variation applies to the way lexical items themselves are assembled. Universal Grammar provides a common store of atomic elements of meaning (such as semantic and grammatical features); languages select and assemble these features into lexical items (LIs). Now, one language may choose to assemble one array of these features into one LI, while another language may assemble a slightly different array into what superficially looks like an equivalent lexical item. A lexical item is nothing but a bundle of features, including phonological, grammatical, and semantic features. This process is illustrated in Figure 2.2, from Gallego (2011: 548).

Let us now recap the main ideas of this section. We discussed the Minimalist view of language variation, according to which most of the differences between languages of the world are encoded in the lexicon. This view is by no means accepted by all generative linguists, see Baker (2008) for one dissenting view. However, this conjecture is at the core of Minimalist theorizing. We also explored three proposals that are not incompatible

¹¹ Evidentiality encodes whether the speaker witnessed the event she is reporting, or it is hearsay.

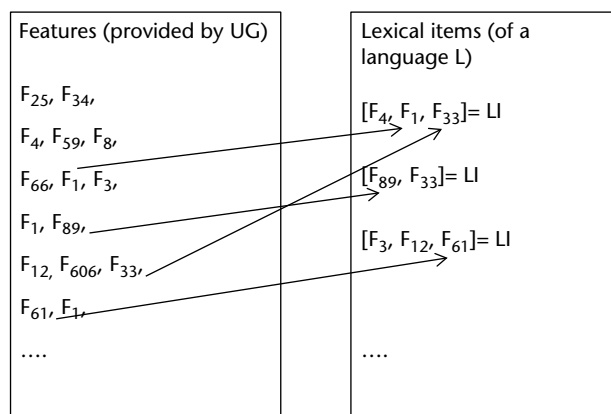


Figure 2.2 Assembling lexical items from features, following Gallego (2011)

with the Borer–Chomsky Conjecture. One proposal was that variation lies at the PF interface, while another proposal was that variation may also lie at the LF interface. Finally, we discussed how the lexicon itself may be a source of variation, through the assembly of different features (phonological, semantic, and grammatical) into actual lexical items of a language.

Why would we care about linguistic proposals on where the differences between languages are located? Particularly if linguists have not made up their own mind just yet? In any cognitive science, current knowledge is in a state of flux. However, we cannot stop research in SLA while waiting for linguists to resolve all their disputes.

Some of the proposals for the locus of linguistic variation make diverging predictions about language acquisition. Think about what these diverging predictions might be.

2.4 What is the learning task for bilinguals?

In order to ponder what the learning task is in L2 acquisition, an important preliminary question that we need to answer refers to the L2 initial state. What do L2 learners bring to the process of L2 acquisition? There is

almost universal consensus¹² that learners do not embark on the task with a clean slate, attempting to activate UG principles and set parameter values solely based on the L2 linguistic experience. They already have a grammar in their head, their native grammar. Following work described in White (1989), Schwartz and Sprouse (1994, 1996) argue that the whole of one's native grammar acts as the initial state of L2 acquisition. This position, known as "L1 transfer," entails that principles are available from UG or from the L1, and one cannot realistically tease these two sources apart. In addition, parameter values are initially transferred from the native language. If the L1 and the L2 have similar parameter values, no adjustment needs to occur; if the L1 and L2 parameter values differ, resetting of parameter values should be accomplished, if the L2 interlanguage grammar is to become targetlike.

While views on parameters in linguistic theory are frequently described,¹³ there have been few attempts (but Lardiere, 2009 is a prominent one, to be discussed at length later on) to analyze what those views on parameters would mean for acquiring a second language after the initial L1 transfer. The way language variation is conceptualized virtually determines what a learner must "do" in order to acquire target language parametric values. The view of L1 transfer described above, enunciated most clearly by Schwartz and Sprouse, is firmly based on the Principles and Parameters framework. As the reader has appreciated in this chapter, generative positions on what parameters are and how they can be acquired have changed radically at least three times over the last 50 or 60 years. However, in the wider second language acquisition community some myths about generative parameters still persist, for example, the myth of parameters as on-and-off switches, and the associated idea of clustering. One of my major goals in this textbook is to show that the "new parameters" have already led to important new lines of research in generative SLA research, and that they hold much promise for the future.

In this section, we will preview some concrete applications of Minimalist parameters that follow logically from what we have discussed in this chapter. In keeping with the Scientific Method, L2 researchers insist on approaching the process of L2 development from the point of view of

¹² However, see Epstein, Flynn, and Martohardjono (1996) for a dissenting view.

¹³ For excellent recent reviews see Biberauer (2008) and Gallego (2011). Holmberg (2010) presents a spirited and convincing defense of parameters.

learning tasks. These tasks spell out what properties have to be acquired in order for the learner to create in her mind the linguistic representations of the target L2 property. One such proposal is the Bottleneck Hypothesis (Slabakova 2008). The gist of this hypothesis is as follows. If language variation is (predominantly) captured in the lexicon, and more specifically in the functional lexicon, then it makes sense to think of language acquisition as acquiring the functional lexicon of the target language. As we mentioned above, this task would subsume learning the whole parametric profile of the new language. That would make it the most important task in language learning. Consequently, this is where learners and teachers have to concentrate their efforts.

The Bottleneck Hypothesis is also interested in what is difficult and what is easy to acquire in a second language. The motivation is again practical: more difficult areas of the grammar should get more instructional effort and attention, as compared to parts of the grammar that need not be taught because they will come to the learner for free. The predictions are based on current views of the language architecture and on the Minimalist proposals on what is to be acquired. Consequently, the hypothesis argues that there is ample evidence for successful acquisition of phrasal semantics and some universal pragmatics, because these properties are universal. It also predicts that (narrow) syntactic computations will not present insurmountable difficulty to learners, because they can transfer these syntactic mechanisms from their native language. On the other hand, acquiring the functional morphology of a second language, together with the related syntactic and semantic effects, should prove to be more difficult.

What would present the gravest difficulty for L2 learners? One of the hardest tasks would be to acquire a mismatch in the exponents of features.¹⁴ This would be the situation when features are scattered across different lexical items. Recall the Spanish–English mismatch where the habitual action meaning is reflected by the simple past tense in English but in the imperfect tense in Spanish. We will see more examples in the exercises of this chapter and later in the book. L2 acquisition is also predicted to be especially difficult when the target parameter values are not fixed in overt functional morphology but signaled by various (lexical, word order) means or left to the discourse to supply. This is so because a specific

¹⁴ As proposed by Lardiere (2009a,b), Ramchand and Svenonius (2008), and Slabakova (2009).

grammatical meaning (say, tense) is not predictably and uniformly signaled by one and the same morpheme every time it needs to be encoded. The learner has to attend to various cues and signals in the surrounding situation and the discourse in order to deduce the meaning. In sum, linguistic-theoretical proposals for parametric differences among languages are still our best tool in investigating second language grammars.

In this chapter, we focused on language variation, discussing several proposals on where the differences between languages may lie. Old-style P&P parameters were visualized as on-and-off switches, bringing clusters of unrelated constructions into the grammar. Minimalist parameters are closely tied to the acquisition of formal features. Mismatches and variation on feature expressions present hazards for successful acquisition.

2.5 Exercises

Exercise 2.1. Kayne (2005: 9) points to the following differences among English, Italian, and French.

- a. We don't know **where** to go.
- b. We don't know **whether** to leave.
- c. *We don't know **if** to leave.
- d. *Jean ne sait pas **si** part-ir (French)
 Jean NEG know-3SG not if leave-INF
- e. Gianni non sa **se** part-ire (Italian)
 Gianni NEG know-3SG if leave-INF

Let us look at the English paradigm first, sentences (a), (b), and (c). The main clause subject (*we*) is the understood subject of the embedded clause. In other words, the people who don't know, and the people who need to go somewhere/leave, are "*we*." These are called control constructions. *Whether* is acceptable as the complementizer of the embedded clause, but *if* is not. (*If* has other requirements for its surroundings, which you may discover by creating your own examples.) Italian allows its equivalent of *if* to appear in structure (e), while French doesn't, as in (d). Discuss whether this is a parametric difference. Can it be described in terms of a binary choice? Is it possible to capture this difference in the functional lexicon of these

languages? Is this difference pervasive in the grammar? Would you change your answer on taking the following additional data into account?

- f. Jean veut l'-achet-er (French)
Jean want-3SG CLITIC-buy-INF
'Jean wants to buy it.'
- g. Gianni vuole compr-ar-lo (Italian)
Gianni want-3SG buy-INF-CLITIC
'Gianni wants to buy it.'

Describe the different orders between object clitic (bold in the examples above) and infinitive in French and Italian. Do you see any connection with the sentences in (d) and (e)? Would English be relevant in the discussion of this difference?

Exercise 2.2. One of the cornerstones of the Principles and Parameters approach was the X-bar schema, which has retained its importance for the theory. It maintains that every phrase in language has the following components: a head of the phrase, a complement, and a specifier. They are in a hierarchical structure as shown below:

- (i)
- XP = the maximal projection

X' = the intermediate projection,
pronounced X-bar

X° = the head

What you have in (i) is the general schema for all languages, but within each language, only one specifier and one complement are permitted. The intermediate levels can be repeated, but each phrase has only one head and only one maximal projection. Keeping the general schema in mind, draw the four logically possible phrase structures. The first one has been provided:

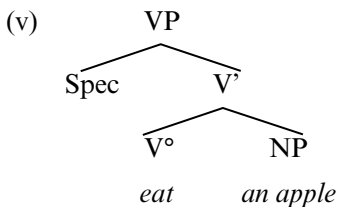
- (ii)
-
- ```
graph TD; XP --> Spec; XP --> X_prime[X']; X_prime --> X_deg[X°]; X_prime --> Complement
```

Next, recall the typological generalizations we made in Chapter 1, repeated here for ease of reference. These pervasive regularities in harmonic languages have been described as the Head Directionality Parameter.

- (iii) If V–O, then Auxiliary–V, Prepositions, Noun–Adjective, Noun–Genitive, Noun–Relative Clause.  
If O–V, then V–Auxiliary, Postpositions, Adjective–Noun, Genitive–Noun, Relative Clause–Noun
- (iv) The head of the phrase precedes/follows its complement.

For the time being, ignore the position of the Specifier (assuming it is always on the left) and focus on the relative positions of the head and its complement. Provide the structures of the following English phrases, where the heads are underlined. The first one is provided.

VP: eat *an apple*; NP: bone *of contention*; AP: proud *of his daughter*; PP: at *the market*



Are English phrases head-initial or head-final? Are all the complements NPs? Does English seem to be entirely harmonic, when you look at the orders in (i)?

Now, what are the predictions that you would make for the equivalent Japanese categories, VP, NP, AP, and PP? Draw their phrase structures.

The X-bar schema is another example of a principle that is also flexible. Describe how the Head Directionality Parameter provides the flexibility of the X-bar schema.

**Exercise 2.3.** This exercise is related to the one above, so do them in order. We have just seen above that the lexical categories (Noun, Verb, Adjective, Preposition) have the structure reflected in (ii). How about the functional heads Determiner, Tense, Aspect, and Complementizer, which project their own functional categories? Do you think they have the same structure? Would it be an acquisitional advantage if they did? Can you think of the

structure of TP in English? What can it have as its complement? Think of examples such as:

- a) John might go to Edinburgh.
- b) John might have been to Edinburgh.
- c) John is going to Edinburgh.

**Exercise 2.4.** Write down concrete predictions for the acquisition of the Null Subject Parameter in English by Italian or Spanish native speakers. What is the English value of this parameter?

Write down concrete predictions for the acquisition of the Compounding Parameter by English-native learners of Spanish. Spanish displays the opposite value of English. What would you expect the learners to know?

**Exercise 2.5.** The White (1985) study of the Null Subject Parameter (also known as the Pro-Drop Parameter) is a classic study in the generative SLA research framework. We mentioned it in Chapter 1. Find it on your library website and download it onto your hard drive. Here is the exact reference.

White, L. 1985. The Pro-Drop Parameter in adult second language acquisition. *Language Learning* Vol. 35, No. 1, pp. 47–62.

Answer the following questions:

1. Which three properties related to the NSP did the study investigate?
2. Why did the researcher include two experimental groups, French-speakers learning English and Spanish-speakers learning English?
3. What kind of test was used? Why were some filler sentences included that were unrelated to the NSP? Were all of the fillers acceptable? Why not?
4. How can you summarize the results? Was the level of proficiency a factor in accepting ungrammatical null subject sentences in English? How about in accepting post-verbal subjects?
5. Were the three NSP constructions acquired as a cluster, according to the findings of this study? Which construction presented the most difficulty?
6. One of the constructions in the purported cluster, the *that*-trace effect, is more complex than the others. The researcher is aware of that discrepancy in complexity. What prediction of the NSP does she

decide to look at in order to avoid this discrepancy (p. 57 of the article).

7. The unacceptability of missing pronominal subjects and of postverbal subjects seem to be of equal complexity, at least as compared to the *that*-trace effect, which is more complex than them. Did the loss of these two properties in the L2 English go hand in hand?
8. What is the author's interpretation of the results? Do you agree with it? Why, or why not?